Programming
Human-Centered Artificial Intelligence
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Exercise sheet 7

Tuples, sets and dictionaries

Not graded, not handed in. Do them anyway. This is where the learning happens! Solutions are published after we discuss them in class.

Ask the question first. Before choosing a structure, write down the question you will ask the data. What is at position i? Is x in there? What belongs to this key? The question picks the structure.

Part A — Tuples and sets

A1 *Eight lines with tuples*

Throughout, `t = (4, 7, 1)`. Give the value of each line, or the error it raises.

```
1.  t[1]                        5.  type((5))
2.  t[-1] + t[0]                6.  a, b = 1, 2
3.  t + (2,)                        a, b = b, a + b     # what are a and b?
4.  len((5,))                   7.  a, b, c = (1, 2)
                                8.  t[0] = 9
```

Number 6 is the one to think about twice. Which side of the second line is worked out first?

A2 *Eight lines with sets*

Throughout, `s = {3, 1, 4, 1, 5}`.

```
1.  len(s)                      5.  s - {1, 5}
2.  1 in s                      6.  len(set("mississippi"))
3.  s | {9}                     7.  type({})
4.  s & {1, 2, 3}               8.  s[0]
```

Explain number 1 in one sentence, and number 7 in another.

A3 *Which structure?*

For each situation, choose a list, a tuple, a set or a dictionary. Justify each choice in one sentence, starting from the question you would ask.

a) The moves of a chess game, in the order they were played.

b) A colour, stored as its red, green and blue values.

c) The countries you have visited, so you can check whether you have been to Japan.

d) A phone book.

e) The number of words in each chapter of a book.

f) The opening hours of a shop for each day of the week.

Number 6 is easiest with two structures, one inside the other.

A4 *The letters a sentence never uses*

Write a program that reads a sentence and prints, in alphabetical order, the letters of the alphabet that do not appear in it. If the sentence uses every letter, it should say so.

a) Write the comment plan first.

b) Check it on these two sentences.

```
the quick brown fox jumps over the lazy dog
Hello, world!
```

c) Spaces and punctuation are in the sentence too. Do they need removing? Why not?

A5 *A list of points*

```
points = [(3, 4), (-1, 2), (0, -5), (6, 1), (-2, -2)]
```

a) Print every point as `x = 3, y = 4`, unpacking each tuple in the loop.

b) Count the points that lie above the horizontal axis.

c) Find the point closest to `(0, 0)`. The distance is $\sqrt{x^2 + y^2}$, and you do not need the square root. Why not?

d) Predict what `sorted(points)` gives, then check.

Part B — Dictionaries

B1 *Ten lines with one dictionary*

```
d = {"a": 1, "b": 2}
d["c"] = 3
d["a"] = 5
```

After these three lines, give the value of each.

```
1.  len(d)                    6.  2 in d
2.  d["a"]                    7.  list(d)
3.  d.get("z")                8.  list(d.values())
4.  d.get("z", 0)             9.  d["z"]
5.  "b" in d                 10.  {"x": 1, "x": 2}
```

B2 *It crashes on the first item*

```
counts = {}
for colour in ["red", "blue", "red"]:
    counts[colour] += 1
print(counts)
```

a) Which error does it raise, and on which pass through the loop?

b) Why? Say what `counts[colour] += 1` asks Python to read first.

c) Fix it with `get`.

d) Fix it again with `in`, without `get`.

B3 *Who won the vote?*

```
votes = ["pizza", "sushi", "pizza", "pasta",
         "sushi", "pasta", "sushi", "pizza"]
```

a) Count the votes for each option and print the counts.

b) Print the winner.

c) Look at your counts again. If your program printed one winner, it is wrong for this data. Make it print every option that has the highest count.

B4 *Word by word*

```
italian = {"the": "il", "cat": "gatto", "eats": "mangia", "fish": "pesce"}
```

Translate `the cat eats the fish quickly` word by word. A word that is not in the dictionary stays as it is. Do it with `get` and no `if`.

Then, in one sentence, say why word by word translation fails for real Italian.

B5 *Turn it round*

a) Turn this dictionary round, so that each capital gives its country.

```
capitals = {"Italy": "Rome", "France": "Paris", "Germany": "Berlin"}
```

b) Now do the same with

```
grades = {"Ada": 30, "Alan": 27, "Grace": 30, "Linus": 27, "Tim": 24}
```

Print the result. Two students have gone missing. Why?

c) Fix it so that each grade gives the *list* of students who got it.

B6 *Student records*

```
records = {
    "Ada":   {"Programming": 30, "Statistics": 28},
    "Alan":  {"Programming": 27},
    "Grace": {"Statistics": 29, "Ethics": 30},
}
```

a) Print Ada's grade in Statistics.

b) Print every student's mean grade, to one decimal place.

c) Print the students who took Programming.

d) Alan gets 26 in Ethics. Add it with one line.

e) A new student, Tim, is added with no grades yet, `records["Tim"] = {}`. What happens to your answer to part 2? Make it print `no grades` instead.

If you finish early

Two words are anagrams if they use the same letters, like `listen` and `silent`. Group these words so that anagrams end up together.

```
words = ["listen", "silent", "enlist", "google",
         "inlets", "banana", "gogole"]
```

Hint: two anagrams give the same result under `sorted`. That result is a list, and a list cannot be a key. Which structure from today can?